

ATM CONTROLLER MACHINE :

The problem statement required designing a **complete ATM controller FSM** that:

1. **Handles user authentication** (PIN verification, block after 3 wrong attempts).
2. **Performs all transaction flows** — cash withdrawal, balance inquiry, and mini statement.
3. **Supports PIN setup/reset** via OTP verification.
4. **Implements cancellation at any point** to safely abort and eject the card.
5. **Checks ATM cash availability** and dynamically displays denominations.

The FSM is also required to be **implemented in Verilog HDL**, verified using a **testbench**, and synthesized to analyze LUT usage, flip-flops, power, latency, and delay

Design Logic and State Transition Algorithm :

The FSM systematically transitions between well-defined states based on user inputs and system conditions, ensuring secure and deterministic ATM operation.

1. Sequential Logic

The sequential logic updates the **current state** and tracks wrong PIN attempts on each positive clock edge.

- On reset, the FSM returns to the **IDLE** state and clears counters.
- When a wrong PIN is entered, a 2-bit counter increments; after three wrong attempts, the card is blocked.

- State transitions occur only at clock edges, ensuring synchronous operation.

2. Combinational Logic

The combinational block determines the **next state** and the **active outputs** based on the current state and input signals.

A global **cancel condition** has the highest priority and can interrupt any process, leading directly to the CANCELLED state, followed by card ejection.

3. Major Functional Flows

- **Cash Withdrawal:**

IDLE → CARD_INSERTED → PIN_ENTRY →
TRANSACTION_MENU → WITHDRAW →
ACCOUNT_TYPE_SELECT → AMOUNT_ENTRY →
CHECK_BALANCE → DENOMINATION_SELECT →
DISPENSE_CASH → RECEIPT_OPTION → CARD_EJECT →
COMPLETE

- **PIN Setup/Reset:**

CARD_INSERTED → ACC_NUMBER_ENTRY → ACCOUNT_VERIFY
→ OTP_SEND → OTP_VERIFY → PIN_CONFIRM →
RECEIPT_OPTION → CARD_EJECT → COMPLETE

- **Balance Inquiry/Mini Statement:**

Simple menu-based flows leading to balance display or statement printing.

4. Error and Cancel Handling

- If invalid inputs or resource shortages occur (!amount_ok, !atm_cash_ok), FSM transitions to the ERROR state and ejects the card.

- Pressing **Cancel** at any stage forces an immediate transition to CANCELLED → CARD_EJECT → COMPLETE → IDLE.
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Analysis of Test bench Simulation :

The provided waveform shows the ATM FSM (atm_controller complete) responding to the sequence of testbench stimulus that exercises eight scenarios (withdrawal, new PIN setup, PIN reset, balance inquiry, mini-statement, cancel, wrong PIN block, and an error path). The state_out bus, control inputs (card, PIN, menu selections, account verification, OTP signals, denomination flags, cancel), and key outputs (dispense_cash, print_receipt, send_otp, display_denom_*, eject_card, display_balance, print_statement) are visible. The clock is a 10 ns period.

Important signals to watch in the waveform

- state_out (5-bit) — shows the current FSM state (numerical codes correspond to named states in the code).
 - Inputs that trigger flows: card_in, pin_entered, pin_valid, menu_*, acc_verified, amount_ok, atm_cash_ok, denom_*_ok, otp_sent, otp_valid, new_pin_set, receipt_req, cancel.
 - Outputs of interest: dispense_cash, print_receipt, eject_card, display_balance, print_statement, send_otp, display_denom_100/500/2000.
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Scenario-by-scenario observations (as seen in waveform)

Scenario 1 — Cash Withdrawal (normal)

- Sequence: card_in pulse → pin_entered=1 with pin_valid=1 → menu_withdraw → acc_type_sel → amount_ok=1 & atm_cash_ok=1 → denom_100_ok, denom_500_ok asserted → receipt_req=1.
- Waveform confirms:
 - state_out transitions through the expected states (IDLE → CARD_INSERTED → PIN_ENTRY → TRANSACTION_MENU → WITHDRAW → ACCOUNT_TYPE_SELECT → AMOUNT_ENTRY → CHECK_BALANCE → DENOMINATION_SELECT → DISPENSE_CASH → RECEIPT_OPTION → CARD_EJECT → COMPLETE → IDLE).
 - During DENOMINATION_SELECT, display_denom_100 and display_denom_500 are asserted (matching denom_100_ok and denom_500_ok inputs).
 - dispense_cash goes high in DISPENSE_CASH.
 - print_receipt goes high in RECEIPT_OPTION because receipt_req=1.
 - eject_card asserted in CARD_EJECT and the FSM returns to IDLE.
- Conclusion: Withdrawal flow behaves correctly — verification, denomination display, dispense and receipt all occur as intended.

Scenario 2 — New PIN Setup (fresh card)

- Sequence: card_in with pin_not_set=1 → ACC_NUMBER_ENTRY → acc_verified=1 → otp_sent=1 → otp_valid=1 → new_pin_set=1 → receipt_req=1.
- Waveform confirms:

- The FSM takes the NEW_PIN_SETUP path and moves to ACCOUNT_VERIFY.
- send_otp becomes active in OTP_SEND (output send_otp asserted), and otp_sent input moves the FSM to OTP_VERIFY.
- After otp_valid=1 and new_pin_set=1, print_receipt is asserted in RECEIPT_OPTION, and the card is ejected.
- Conclusion: PIN setup flow (including account verification, OTP send/verify, and PIN confirmation) is correctly modeled.

Scenario 3 — PIN Reset (existing card)

- Sequence: Normal PIN entry → menu_pinchange → ACC_NUMBER_ENTRY → acc_verified=1 → otp_sent=1 & otp_valid=1 → new_pin_set=1 → optional receipt.
- Waveform confirms identical OTP-based flow as Scenario 2 for PIN reset; send_otp, otp_sent, otp_valid, new_pin_set, and print_receipt behave as expected.
- Conclusion: PIN reset flow works correctly.

Scenario 4 — Balance Inquiry

- Sequence: Card + valid PIN → menu_balance=1 → display_balance asserted and print_receipt optionally asserted.
- Waveform confirms display_balance goes high in the BALANCE_INQUIRY state and print_receipt follows if requested.
- Conclusion: Balance inquiry behavior is correct.

Scenario 5 — Mini Statement

- Sequence: Card + valid PIN → menu_statement=1 → print_statement asserted → card ejected.

- Waveform confirms print_statement goes high and the FSM ejects the card afterward.

Scenario 6 — Cancel during operation

- Sequence: Start withdrawal flow, then assert cancel=1.
- Waveform confirms:
 - On cancel assertion the FSM immediately transitions to CANCELLED (global override), eject_card asserted, and moves to COMPLETE then IDLE.
 - No dispense_cash or print_receipt occur after cancel — transaction aborted safely.
- Conclusion: Cancel has top priority and correctly aborts any active operation.

Scenario 7 — Wrong PIN Block

- Sequence: card_in then three repeats of pin_entered=1 with pin_valid=0.
- Waveform confirms:
 - Repeated wrong entries keep state_out in PIN_ENTRY until wrong_pin_count reaches 3 and then FSM transitions to BLOCKED.
 - eject_card asserted and system goes to COMPLETE.
- Conclusion: Wrong-PIN counting and block behavior appear correct. (The internal counter is inferred via state behavior; to observe it explicitly, instrument the design or add a waveform signal.)

Scenario 8 — ERROR path (ATM out of cash)

- Sequence: Withdrawal with amount_ok=1 but atm_cash_ok=0.

- Waveform confirms:
 - At CHECK_BALANCE the FSM sees !atm_cash_ok and moves to ERROR.
 - eject_card asserted from ERROR and transaction ends.
- Conclusion: Error handling for insufficient ATM cash works as intended

Timing and waveform-level observations

- card_in is applied as a pulse (#10 card_in = 1; #10 card_in = 0) — this cleanly triggers a single insertion event.
- Outputs such as dispense_cash, send_otp, print_receipt, and eject_card appear as pulses tied to their respective states (asserted combinationally while the FSM is in that state).
- display_denom_* lines reflect the provided denom_*_ok inputs during DENOMINATION_SELECT, which validates the dynamic denomination display logic.
- state_out values change synchronously on clock edges and align with input events delayed by one clock cycle (as expected in a synchronous FSM).

Hardware Synthesis and Performance Analysis :

1.LUT and Flip-Flop Utilization Report:

The purpose of this section is to analyze the FPGA resource utilization, particularly focusing on Look-Up Tables (LUTs) and Flip-Flops (FFs) used in implementing the ATM Controller FSM on the

ZedBoard (XC7Z020-CLG484) FPGA device.

Site Type	Used	Fixed	Prohibited	Available	Util%
Slice LUTs*	42	0	0	53200	0.08
LUT as Logic	42	0	0	53200	0.08
LUT as Memory	0	0	0	17400	0.00
Slice Registers	7	0	0	106400	<0.01
Register as Flip Flop	7	0	0	106400	<0.01
Register as Latch	0	0	0	106400	0.00
F7 Muxes	1	0	0	26600	<0.01
F8 Muxes	0	0	0	13300	0.00

Resource Type	Used	Available	Utilization (%)	Description
Slice LUTs	42	53,200	0.08%	Used for combinational logic (FSM next-state and output logic).
LUT as Logic	42	53,200	0.08%	Indicates all LUTs are used purely for logic computation, not memory.
LUT as Memory	0	17,400	0.00%	No LUTs configured as distributed RAM or shift registers.
Slice Registers	7	106,400	<0.01%	Used for sequential elements like FSM state registers and counters.
Registers as Flip-Flops	7	106,400	<0.01%	Indicates flip-flops are used for synchronous state storage.
Registers as Latches	0	106,400	0.00%	Confirms purely synchronous design — no transparent latches used.
F7 Muxes	1	26,600	<0.01%	Used for LUT output selection or multiplexing.
F8 Muxes	0	13,300	0.00%	No higher-level multiplexing logic used.

2.Power Analysis Report :

The purpose of this analysis is to evaluate the on-chip power consumption of the *ATM Controller FSM* implemented on the ZedBoard (XC7Z020-CLG484) device.

Power analysis provides insight into the dynamic and static components of power usage, helping verify the design's efficiency and thermal safety.

Parameter	Reported Value	Description / Interpretation
Total On-Chip Power	4.942 W	Total power consumed by the FPGA, including dynamic and static components.
Dynamic Power	4.636 W	Power consumed due to switching activities in logic, routing, and I/O during operation.
Device Static Power	0.306 W	Power consumed by leakage currents and biasing when the FPGA is powered but idle.

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Total On-Chip Power (W)	4.942	
Design Power Budget (W)	Unspecified*	
Power Budget Margin (W)	NA	
Dynamic (W)	4.636	
Device Static (W)	0.306	
Effective TJA (C/W)	11.5	
Max Ambient (C)	28.0	
Junction Temperature (C)	82.0	
Confidence Level	Low	
Setting File	---	
Simulation Activity File	---	
Design Nets Matched	NA	

3.Timing and Performance Analysis Report :

The objective of this analysis is to determine the critical path delay, latency, and maximum achievable operating frequency of the *ATM Controller FSM* implemented on the ZedBoard (XC7Z020-CLG484) FPGA.

Timing analysis ensures that the design operates synchronously without timing violations and meets real-time system constraints.

Parameter	Reported Value	Description / Interpretation
Critical Path Delay (Tpd)	8.301 ns	The longest combinational delay between an input (cancel) and an output (eject_card), determining system speed.
Logic Delay	3.737 ns (45%)	Delay due to logic elements (LUT6 and flip-flops).

Parameter	Reported Value	Description / Interpretation
Routing Delay	4.564 ns (55%)	Delay caused by interconnect wiring and routing on FPGA fabric.
Total Path Levels	3 levels	IBUF → LUT6 → OBUF — representing a single FSM output transition path.
Minimum Path Delay	0.357 ns	Shortest propagation delay (internal register-to-register path).
Maximum Operating Frequency (f_{max})	≈ 120.4 MHz	Computed as $f_{\max} = 1 / T_{pd} = 1 / 8.301 \times 10^{-9} \text{ s}$.
Timing Slack	∞ (No Violations)	Indicates the design meets all timing constraints successfully.
Latency	One clock cycle (~8.3 ns)	FSM transitions occur per clock cycle; total latency = one state transition per rising edge.